Kaixun Hua

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PROFESSIONAL EXPERIENCE

University of South Florida, Tampa, FL — 08/2023 – Current
Assistant Professor, Department of Industrial and Management Systems Engineering
Research Expertise: Interpretable Machine Learning and Large-scale Global Optimization

University of British Columbia, Vancouver, BC — 07/2020 – 05/2023
Postdoctoral Fellow, Institute of Applied Mathematics — Advisor: Yankai Cao
Research Topic: Next Generation Training Algorithms with Deterministic Global Optimality

University of Massachusetts Boston, Boston, MA — 09/2013 – 05-2019
Graduate Research Assistant

EDUCATION

University of Massachusetts Boston, Boston, MA — 09/2013 – 05/2019
Ph.D., Computer Science — Advisor: Dan A. Simovici
Thesis: Clusterability, Model Selection and Evaluation

Cornell University, Ithaca, NY — 09/2012 – 08/2013
M.Eng., Systems Engineering — Advisor: Hsiao-Dong Chiang
Thesis: A Study on State Estimation of Power System with Uniform Distributed Residuals

Shanghai Jiao Tong University, Shanghai, China — 09/2008 – 08/2012
B.Sci., Electrical and Computer Engineering

HONORS & AWARDS

- INFORMS 2025 Data Mining Best Paper Award **Runner-up** (Theoretical track)
- INFORMS 2025 Minority Issues Forum (MIF) Poster Competition **Honorable Mention**
- NeurIPS 2022 Scholar Award (Total Award Amount: $3,200)
- INFORMS 2022 Data Mining Best Paper Award **Winner** (Theoretical track)
- PICS Scholar (Pacific Institute for Climate Solutions)
- ICML 2022 Participation Grant (Total Award Amount: $2,850)
- Best Presentation Award (4th BC Universities Systems and Control Meeting, 2021)
- Outstanding Research Award (Doctoral Dissertation, UMass Boston)
- Annual Research Symposium Competition (1st Place '18, 3rd Place '16, UMass Boston)
- The Third Prize Scholarship (2011-2012, Shanghai Jiao Tong University)
- Dean's List (2010-2012, University of Michigan-Shanghai Jiao Tong University Joint Institute)

CONFERENCE PUBLICATIONS

1. [**NeurIPS 2025**] Xiong, X, Flores, P. C., **Hua, K.**, & Hua, C. (2025). SPOT: Scalable Policy Optimization with Trees for Markov Decision Processes. *Advances in Neural Information Processing Systems.* Accepted. (**First-Corresponding Author**).
2. [**IEEE BigData 2025 Long Paper**] Chumpitaz-Flores, P., Duong, M., Mao, Y., & **Hua, K.** (2025). qc-kmeans: A Quantum Compressive K-Means Algorithm for NISQ Devices. arXiv preprint arXiv:2510.22540. *IEEE International Conference on Big Data.* Accepted. (**Corresponding Author**).
3. [**IEEE BigData 2025**] Chumpitaz-Flores, P., Duong, M., Mao, Y., & **Hua, K.** (2025). QIBONN: A Quantum-Inspired Bilevel Optimizer for Neural Networks on Tabular Classification. *IEEE International Conference on Big Data.* Accepted. (**Corresponding Author**).

4. **[ECAI 2023]** Liu, K. Qiang, J. Li, Y. Yuan, Y. Zhu, Y. & **Hua, K.** (2023). Multilingual Lexical Simplification via Paraphrase Generation. *European Conference on Artificial Intelligence* (pp. 1529-1535).
5. **[NeurIPS 2022] Hua, K.**, Ren, J., & Cao, Y. (2022). A Scalable Deterministic Global Optimization Algorithm for Training Optimal Decision Tree. *Advances in Neural Information Processing Systems*, 35, 8347-8359.
6. **[NeurIPS 2022]** Ren, J.*, **Hua, K.***, & Cao, Y. (2022). Global Optimal K-Medoids Clustering of One Million Samples. *Advances in Neural Information Processing Systems*, 35, 982-994. (**Co-first Author**)
7. **[ICML 2022 Spotlight]** Shi, M.*, **Hua, K.***, Ren, J., & Cao, Y. (2022). Global Optimization of K-Center Clustering. In *International Conference on Machine Learning* (pp. 19956-19966). PMLR. (**Co-first Author**)
8. **[ICML 2021] Hua, K.**, Shi, M., & Cao, Y. (2021). A Scalable Deterministic Global Optimization Algorithm for Clustering Problems. In *International Conference on Machine Learning* (pp. 4391-4401). PMLR.
9. **[MMCTSE 2019]** Simovici, D. A. & **Hua, K.** (2019). Data Ultrametricity and Clusterability. In *Journal of Physics: Conference Series* (Vol. 1334, No. 1, p. 012002). IOP Publishing.
10. **[SYNASC 2018] Hua, K.**, & Simovici, D. A. (2018). Dual Criteria Determination of the Number of Clusters in Data. In *2018 20th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing* (pp. 201-208). IEEE.
11. **[ICNSC 2016] Hua, K.**, & Simovici, D. A. (2016). Long-lead Term Precipitation Forecasting by Hierarchical Clustering-based Bayesian Structural Vector Autoregression. In *2016 IEEE 13th International Conference on Networking, Sensing, and Control* (pp. 1-6). IEEE.

JOURNAL PUBLICATIONS

1. Palma, M., Kan, S., Wei, W., Chen, J., **Hua, K**, Mouradian, S., & Mao, Y. (2025). Hardware-aware and Resource-efficient Circuit Packing and Scheduling on Trapped-Ion Quantum Computers. ***IEEE Transactions on Quantum Engineering***.
2. Ren, J., You, N., **Hua, K.**, Ji, C., & Cao, Y. (2025). A Global Optimization Algorithm for K-Center Clustering of One Billion Samples. ***Management Science***.
3. Liu, K., Qiang, J., Li, Y., Zhu, Y., Yuan, Y., **Hua, K.**, & Wu, X. (2025). Multilingual Lexical Simplification via Zero-Shot Paraphrasing. ***IEEE Transactions on Audio, Speech and Language Processing***.
4. Savachkin, A., Ardila-Rueda, W., Romero-Rodriguez, D., **Hua, K.**, & Cruz, J. N. D. L. (2025). Assessment and Optimization of System Resilience: A Critical Review. ***Models and Methods for Systems Engineering***, 251-260.
5. Ren, J., **Hua, K.**, Trajano, H., & Cao, Y. (2023). Global Optimal Explainable Models for Biorefining. ***Computer Aided Chemical Engineering*** (Vol. 52, pp. 1339-1346). Elsevier.
6. Li, Y., **Hua, K.**, & Cao, Y. (2022). Using Stochastic Programming to Train Neural Network Approximation of Nonlinear MPC Laws. ***Automatica***, 146, 110665.
7. Li, Y., Wang, Y., Chen, Y., Lu, Y., **Hua, K.**, Ren, J., Mozafari, G., Lu, Q. & Cao, Y., (2022). Deep-Learning-Based Predictive Control of Battery Management for Frequency Regulation. ***Industrial & Engineering Chemistry Research***, 61(24), 8432-8442.
8. Chen, Q., Zuo, L., Wu, C., Li, Y., **Hua, K.**, Mehrtash, M., & Cao, Y. (2022). Optimization of compressor standby schemes for gas transmission pipeline systems based on gas delivery reliability. ***Reliability Engineering & System Safety***, 221, 108351.
9. Mehrtash, M., Mozafari, G., **Hua, K.**, & Cao, Y. (2022). Stochastic Optimal Device Sizing Model for Zero Energy Buildings: A Parallel Computing Solution. ***IEEE Transactions on Industry Applications***, 58(3), 3275-3284.
10. Xu, Z., Lian, J., Wang, R., Qiu, Y., Song, T., & **Hua, K.** (2022). Development of Method for Assessing Water Footprint Sustainability. ***Water***, 14(5), 694.(**Co-Corresponding Author**)

11. Xu, Z., Lian, J., Bin, L., **Hua, K.**, Xu, K., & Chan, H. Y. (2019). Water Price Prediction for Increasing Market Efficiency Using Random Forest Regression: A Case Study in the Western United States. ***Water***, 11(2), 228. (**Co-Corresponding Author**).
12. Simovici, D. A., Vetro, R., & **Hua, K.** (2017). Ultrametricity of dissimilarity spaces and its significance for data mining. ***Advances in Knowledge Discovery and Management*** (pp. 141-155). Springer, Cham.
13. Scheffner, I., **Hua, K.**, Simovici, D., Abeling, T., Haller, H., & Gwinner, W. (2016). Prediction of Patient Survival After Kidney Transplantation: Construction, Validation and Evaluation of Decision Models Using Data Mining Approaches. ***American Journal of Transplantation*** (Vol. 16, pp. 572-573).

SELECTED ORAL PRESENTATIONS

1. **Hua, K.**, Ren, J., & Cao, Y. "A Scalable Deterministic Global Optimization Algorithm for Training Optimal Decision Tree." UBC Postdoc Research Day, Vancouver, BC, December 2022.
2. **Hua, K.**, Ren, J., & Cao, Y. "A Scalable Deterministic Global Optimization Algorithm for Training Optimal Decision Tree." The 36th Conference on Neural Information Processing Systems (NeurIPS 2022), New Orleans, LA, November 2022.
3. Ren, J., **Hua, K.**, & Cao, Y. "Global Optimal K-Medoids Clustering of One Million Samples." The 36th Conference on Neural Information Processing Systems (NeurIPS 2022), New Orleans, LA, November 2022.
4. Ren, J., **Hua, K.**, & Cao, Y. "A Scalable Global Optimization Algorithm for K-Medoids Clustering." The 72nd Canadian Chemical Engineering Conference (CCEC 2022), Vancouver, BC, October 2022.
5. **Hua, K.**, Ren, J., & Cao, Y. "A Deterministic Global Optimization Algorithm for Training Optimal Decision Tree on Large Datasets." The 17th INFORMS Workshop on Data Mining and Decision Analytics, Indianapolis, IN, October 2022 (**INFORMS Data Mining Best Theoretical Paper Award**).
6. Shi, M., **Hua, K.**, Ren, J., & Cao, Y. "Global Optimization of K-Center Clustering." Spotlight, The 39th International Conference on Machine Learning (ICML 2022), Baltimore, MD, July 2022. (Virtual)
7. **Hua, K.** "Trustworthy Machine Learning and Its Recent Progress." CS105, Guest Lecture, University of Massachusetts Boston, Boston, MA, December 2021. (Virtual)
8. **Hua, K.**, Mozafari, G. & Cao, Y. "Tutorial for Julia and JuMP." BC Hydro, Vancouver, BC. November 2021. (Virtual)
9. **Hua, K.** & Cao, Y. "A Scalable Global Optimization Algorithm for Stochastic Nonlinear Programs." Conference of the International Federation of Operational Research Societies (IFORS 2021), Seoul, South Korea, August 2021. (Virtual)
10. **Hua, K.** & Cao, Y. "A Scalable Deterministic Global Optimization Algorithm for Clustering Problems." 4th BC Universities Systems and Control Meeting. University of Victoria, Victoria, BC, August 2021. (**Best Presentation Award**) (Virtual)
11. **Hua, K.** & Cao, Y. "A Scalable Deterministic Global Optimization Algorithm for Clustering Problems." SIAM Conference on Optimization (OP21), Spokane, WA, USA, July 2021. (Virtual)
12. **Hua, K.**, Shi, M., & Cao, Y. "A Scalable Deterministic Global Optimization Algorithm for Clustering Problems." Spotlight, The 38th International Conference on Machine Learning (ICML 2021), Vienna, Austria, July 2021. (Virtual)
13. **Hua, K.** "Improving Auto-parts Production Process with Large-scale Machine Learning Algorithm." Ningbo TIJ Automotive Parts Co., Ltd, Zhejiang, China. June 2021.
14. **Hua, K.** & Cao, Y. "A Global Optimization Algorithm for Clustering Problem." Canadian Mathematical Society, Montreal, QC, December 2020. (Virtual)
15. Simovici, D. A. & **Hua, K.**"Clusterability and Model Selection." Bigwood System Inc., Cornell Business & Technology Park, Ithaca, NY. August 2019.
16. **Hua, K.** "Application of Machine Learning in Power System Industry." State Grid Corporation of China, Shanghai, China. July 2018.

17. **Hua, K.** "Data Ultrametricity and Clusterability." Annual Research Symposium, Department of Computer Science, University of Massachusetts Boston, Boston, MA, May 2018.
18. **Hua, K.** "On Finding Natural Clustering Structures in Data." Annual Research Symposium, Department of Computer Science, University of Massachusetts Boston, Boston, MA, May 2016.

TEACHING EXPERIENCE

- *Instructor: ESI4607/6613 Analytics II: Applied Data Science*
  University of South Florida, Tampa, Fl — Spring 2025, 2026
- *Instructor: ESI6681 Deep Learning Analytics*
  University of South Florida, Tampa, Fl — Spring 2024, 2025, Fall 2025
- *Instructor: EIN4933/6935 Foundations of Machine Intelligence*
  University of South Florida, Tampa, Fl — Fall 2024
- *Teaching Assistant: CS110 Introduction to Computing (Python)*
  University of Massachusetts Boston, Boston, MA — Fall & Spring 2015 - 2019
- *Teaching Assistant: CS420 Introduction to the Theory of Computation*
  University of Massachusetts Boston, Boston, MA — Fall 2014, Spring 2015
- *Teaching Assistant: CS240 Programming in C*
  University of Massachusetts Boston, Boston, MA — Fall 2013, Spring 2014
- *Teaching Assistant: VE311 Electronic Circuits*
  Shanghai Jiao Tong University, Shanghai, China — Summer 2012

RESEARCH MENTORING

- **Doctoral Students**:
  - Cristbal Heredia Galleguillos, University of South Florida, 2024 - present
  - Pedro Roberto Chumpitaz Flores, University of South Florida, 2024 - present
  - Jose Navarro De La Cruz (co-advised with Alex Savachkin), University of South Florida, 2024 - present
- **Master Students**:
  - Pufan You, Cornell University, 2025 - present
  - Yulu Dai, University of Washington, 2025 - present
  - Sweta Jaishankar Ratnani, University of South Florida, 2024 - present
  - Yitian Lou, University of Waterloo, 2024 - present
  - Srivalli Chintapalli, University of South Florida, 2024 - 2025
  - Mingfei Shi, University of British Columbia, 2020 - 2022
- **Undergraduate Students**:
  - Xuyuan Xiong, Shanghai Jiao Tong University, 2025 - present
  - Carlos Aleixo De Lima Domingues, University of South Florida, 2024 - present
  - My Duong, University of South Florida, 2025 - present
  - Alimzhan Kussainov, University of South Florida, 2023 - present
  - Shraman Pal, University of British Columbia/IIT Kharagpur (India), 2022

SCIENTIFIC COMMUNITY AND OUTREACH ACTIVITIES

- Manuscript Review:
  - International Conference on Machine Learning (ICML)
  - Conference on Neural Information Processing Systems (NeurIPS)

- International Conference on Learning Representations (ICLR)
- International Conference on Knowledge Discovery and Data Mining (KDD)
- IEEE International Conference on Data Mining (ICDM)
- IEEE International Conference on Big Data (ICBD)
- The Conference on Information and Knowledge Management (CIKM)
- European Conference on Artificial Intelligence (ECAI)
- IEEE International Conference on Distributed Computing Systems (ICDCS)
- Transactions on Knowledge Discovery from Data
- Knowledge and Information System
- Artificial Intelligence
- Annals of Mathematics and Artificial Intelligence
- Automatica
- Connection Science
- IEEE Internet of Things Journal
- ACM Transactions on Multimedia Computing, Communications, and Applications

- Proposal Reviewer/Panelist:
  - NSF Panelist
- Conference Organization:
  - Session Chair, *Foundations*, IEEE International Conference on Data Mining (ICDM), 2025
  - Student Volunteer Chair, IEEE International Conference on Data Mining (ICDM), 2025
  - Workshop Co-chair, *Workshop on Big Data and AI for Healthcare*, IEEE International Conference on Big Data (IEEE BigData), 2024
  - Session Chair, *Artificial Intelligence and Machine Learning in Process Systems Engineering*, Canadian Chemical Engineering Conference (CCEC), 2022
  - Session Chair, *Data Driven Analytics, Controls, and Optimization (Tuesday AM2)*, Canadian Chemical Engineering Conference (CCEC), 2022
  - Session Chair, *Data Driven Analytics, Controls, and Optimization (Wednesday AM2)*, Canadian Chemical Engineering Conference (CCEC), 2022
- Member Affiliation:
  - Association for Computing Machinery (ACM), Since 2021
  - Society for Industrial and Applied Mathematics (SIAM), Since 2021
  - The Institute for Operations Research and the Management Sciences (INFORMS), Since 2021
  - Association for Information Systems (AIS), Since 2021
  - Institute of Electrical and Electronics Engineers (IEEE), Since 2016